

1. Administrative & Study Identification

**Full Study Title:** An Open-label, Phase 1, Single Ascending Dose-Finding Study to Characterize the Safety, Tolerability, and Pharmacokinetics of a Long Acting Injectable KarXT Formulation in Participants With Schizophrenia  
**Short Title/Acronym:** Single KarXT Intramuscular Injection in Participants With Schizophrenia  
**Protocol Number:** CN012-0016  
**NCT Number:** NCT07061288  
**Sponsor Name:** Bristol-Myers Squibb  
**Investigational Product:** KarXT Long-Acting Injectable (Intramuscular Injection)  
**Phase of Development:** Phase 1  
**Medical Monitor:** BMS Clinical Trials Contact Center (clinical.trials@bms.com)

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2. Study Synopsis & Rationale

**2.1 Study Objective** **Primary Objective:** To evaluate the safety, tolerability, and pharmacokinetics (drug levels) of a single ascending dose of KarXT intramuscular injection in participants with schizophrenia. **Secondary Objectives:** To understand how the body processes the medication over time to determine the optimal safe and effective dosage for managing symptoms.

**2.2 Background & Rationale** To address the need for a safe, well-tolerated, long-acting treatment option for individuals diagnosed with schizophrenia. A long-acting injectable formulation of KarXT could potentially lead to better medication adherence, optimized drug processing, and improved quality of life for those managing schizophrenia symptoms.

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3. Study Design & Methodology

**3.1 Design Framework** **Study Type:** Interventional **Control Type:** Single-arm (Single Ascending Dose-Finding) **Blinding:** Open-label **Randomization:** N/A

**3.2 Planned Enrollment & Duration** **Target N\$:** [Not specified in registry brief, Phase 1 typical cohort size] **Number of Sites:** 8 Locations **Estimated Study Start:** 03-Sep-2025 **Estimated Primary Completion:** [To be determined/Ongoing]

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## 4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Male and female participants aged 18 to 55 years. 2. Primary diagnosis of schizophrenia, confirmed by psychiatric evaluation based on DSM-5-TR criteria and MINI (version 7.0.2). 3. Positive and Negative Syndrome Scale (PANSS) total score  $\leq 80$  and a Clinical Global Impression-Severity (CGI-S) score  $\leq 4$  at both screening and baseline. 4. Body mass index (BMI) between 18 and 40 kg/m<sup>2</sup>.

**4.2 Exclusion Criteria** 1. Newly diagnosed schizophrenia or a first treated episode of schizophrenia. 2. Any other DSM-5-TR disorder diagnosed within the past 12 months, such as major depressive disorder or bipolar disorder. 3. History of alcohol or drug use disorder within the past 12 months, or at risk for suicidal behavior. 4. Female participants who are pregnant or breastfeeding.

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## 5. Intervention & Treatment Plan

**5.1 Investigational Regimen Dosage/Frequency:** Single ascending dose **Route of Administration:** Intramuscular (IM) injection **Duration of Treatment:** Single injection followed by observational safety and pharmacokinetic monitoring period

**5.2 Concomitant & Prohibited Therapy Permitted:** Standard allowed therapies per protocol. **Prohibited:** Participants must be willing and able to discontinue all other antipsychotic medications prior to the baseline visit.

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## 6. Study Timeline & Visit Schedule

| Visit ID     | Window (Days) | Key Activities/Procedures                                                         |
|--------------|---------------|-----------------------------------------------------------------------------------|
| Screening    | Up to Day -1  | Informed Consent, Psychiatric Evaluation (MINI, PANSS, CGI-S), Medical History    |
| Baseline     | Day 0         | Discontinuation of prior antipsychotics, Vital Signs, Quality Audit               |
| Intervention | Day 1         | Administration of single KarXT intramuscular injection                            |
| Follow-up    | Post-Dose     | Safety monitoring, blood draws for pharmacokinetics (drug levels), AE observation |

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## 7. Outcome Measures (Endpoints)

**7.1 Primary Endpoint Definition:** Safety, tolerability, and pharmacokinetic profile (drug concentration levels over time) following a single KarXT intramuscular injection. **Measurement Method:** Blood sampling for PK analysis, continuous monitoring for adverse events, and clinical observation.

**7.2 Secondary & Exploratory Endpoints** Incidence and severity of Treatment-Emergent Adverse Events (TEAEs) and Serious Adverse Events (SAEs). Characterization of the optimal safe and effective dosage for future Phase 2/3 trials.

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## 8. Safety & Adverse Event Reporting

**Adverse Event (AE) Monitoring:** Intensive monitoring of participants to observe how their bodies respond to the treatment, focusing on side effects or changes in health status. **Serious Adverse Events (SAEs):** Standard protocol; reported typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review committee overseeing dose-escalation decisions.

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Safety Set (all participants receiving at least one dose of KarXT), Pharmacokinetic (PK) Set. **Sample Size Justification:** Standard Phase 1 single ascending dose (SAD) powering, sufficient to characterize PK profile and identify dose-limiting toxicities. **Handling of Missing Data:** PK parameters calculated using available data; missing safety data handled per standard early-phase conventions.

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite monitoring for source data verification, PK sample handling, and safety reporting. **Audit Trail:** eCRF locking and comprehensive data clarification mechanisms.

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## 11. Study Identifiers & Governance

**Primary ID:** CN012-0016 **Registry Number:** NCT07061288 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** A Study to Evaluate the Dose Levels, Safety, and Drug Levels of Single KarXT Intramuscular Injection in Participants With Schizophrenia **Official Regulatory Title:** An Open-label, Phase 1, Single Ascending Dose-Finding Study to Characterize the Safety, Tolerability, and Pharmacokinetics of a Long Acting Injectable KarXT Formulation in Participants With Schizophrenia

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## 12. Clinical Framework (Schizophrenia Specific)

**Primary Condition:** Schizophrenia **Diagnosis Threshold:** Primary diagnosis of schizophrenia via DSM-5-TR & MINI; PANSS  $\leq 80$ , CGI-S  $\leq 4$  **Medical Methodology:** Discontinuation of oral antipsychotics followed by novel therapeutic administration **Optimization Target:** Establishment of safe PK profile for long-acting injection

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## 13. Study Procedures & Methodology

**Management Pathway:** In-patient/Out-patient observation for early-phase PK tracking **Education Protocol (HCP):** Investigator training on IM injection protocol and AE monitoring **Education Protocol (Patient):** Informed consent detailing expected side effects and washout requirements **Quality Audit Mechanism:** Close observation of side effects and serious adverse events tracking

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Protocol-defined follow-up weeks for PK clearance **Onsite Visit Cadence:** Intensive early days post-injection for PK sampling **Remote Monitoring Frequency:** Routine post-discharge follow-up **Checklist Review Interval:** Continuous throughout the SAD cohort evaluation

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|------|-----|-----|
| Role                     | Name   | Signature | Date          | :-- | :--  | :-- | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     | Lead |     |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |      |     |     |